

Elementary row operations

To solve a linear system one performs a series of algebraic operations:

- Interchange equations
- Multiply an equation by a non-zero constant
- Add a constant multiple of one equation to another

The main fact is that doing these operations doesn't change the set of solutions of the linear system.

The corresponding *elementary row operations* on an augmented matrix are:

- Interchange two rows
- Multiply a row by a non-zero constant
- Add a constant multiple of one row to another

Ex: Solve

$$y + 3z = 4$$

$$2x - y + z = 2$$

Aug. matrix: $\left(\begin{array}{ccc|c} 0 & 1 & 3 & 4 \\ 2 & -1 & 1 & 2 \end{array} \right)$

$\leftarrow \begin{matrix} 2 \times 4 \\ 2 \times 2 \end{matrix}$

$\underbrace{\quad}_{\text{coeff matrix } 2 \times 3} \quad \underbrace{\quad}_{\text{RHS } 2 \times 1}$

Apply elem. oper.
to this matrix:
row echelon
form.

$\rightarrow \left(\begin{array}{ccc|c} \textcircled{2} & -1 & 1 & 2 \\ 0 & \textcircled{1} & 3 & 4 \end{array} \right)$

$\uparrow \quad \uparrow$
pivot pivot

Divide by 2:

$$\left(\begin{array}{cccc} 1 & -\frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 1 & 3 & 4 \end{array} \right)$$

$$\frac{1}{2} \times \text{row 2} + \text{row 1}$$

$$\left(\begin{array}{cccc} 1 & 0 & 2 & 3 \\ 0 & 1 & 3 & 4 \end{array} \right)$$

reduced
row
echelon form

$$\begin{array}{rcl} x - 2z & = & 3 \\ y + 3z & = & 4 \end{array}$$

$\uparrow$
z is free.

$$y = 4 - 3z.$$

$$\begin{array}{l} x = 3 - 2z \\ y = 4 - 3z \\ z = t \end{array} \text{ for any } t.$$

Ex:

1	2	-3	1
0	1	0	-1
0	0	1	2

↑
row-echelon
form.

Thinking of this
as an aug.
matrix. there
is exactly one
solution.

$$z = 2$$

$$y = -1$$

$$x = -2y + 3z + 1 \\ = 9.$$

① Augmented matrix looks like

$$\left(\begin{array}{ccc|cc} \textcircled{1} & \textcircled{2} & \textcircled{-1} & \textcircled{5} & \textcircled{2} \\ 0 & 0 & \textcircled{1} & \textcircled{2} & \textcircled{-1} \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \leftarrow \text{all zeros.}$$

$\nwarrow$ pivot
columns.

The second and fourth variables will be free in the general solution.

Ex:

$$\left(\begin{array}{cccc} \textcircled{1} & 0 & 1 & 2 \\ 0 & 0 & \textcircled{1} & 4 \\ 0 & 0 & 0 & \textcircled{1} \end{array} \right)$$

← row-echelon form.

As an augmented matrix?

No solution! The third equation can't be solved.

Echelon forms

A matrix is in **row echelon form** if

- 1 All zero rows are grouped contiguously at the bottom of the matrix.
- 2 In any two consecutive rows which are both non-zero, the non-zero entry of the upper row is to the left of the non-zero entry of the lower row.
- 3 Note: Many presentations require that the left most non-zero entry in each row of a non-zero row be a 1. Your book does not demand this.

Reduced row echelon form

If a matrix is in row echelon form then we say the matrix is in **reduced row echelon form** if any pivot column contains a 1 (called a **leading 1**) and has zeroes elsewhere.

Theorem

Every matrix can be row reduced to a unique matrix in reduced row echelon form.


$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The row reduction procedure

- 1 Given a matrix, find the left most column which is not all zero.
- 2 Interchange rows so that the non-zero entry from the previous step is the top row.
- 3 If the non-zero entry you have found is a then divide the top row by a so that its left most entry is 1.
- 4 Work down row by row adding multiples of the first row to each row to guarantee that all entries below this 1 are zero.
- 5 Now ignore the top row and repeat the process with the rows you have remaining until there are no more rows left; this gets the matrix into row echelon form with leading 1's in pivot columns.
- 6 For reduced row echelon form, start with the right most leading 1 and working left, add suitable multiples of that row to the rows above to guarantee that all entries above leading 1's are 0.

Solving systems of linear equations using row reduction

- 1 Start with a linear system and form the augmented matrix.
- 2 Row reduce this matrix to its reduced row echelon form.
- 3 If there is a leading 1 in the right most column, the system has no solutions. Otherwise:
- 4 Every column that is not a pivot column corresponds to a **free variable**; variables corresponding to pivot columns are called **basic variables**.
- 5 Solve the original system by writing the basic variables in terms of the free variables. This gives the general solution to the original system.